ASHEN: Automated Surveillance and Hostile Intent Evaluation Network — A Multimodal Transformer Framework for Weakly Supervised Video Anomaly Detection

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Abstract

Video anomaly detection (VAD) in surveillance systems is a critical task for public safety, yet it remains challenging due to the scarcity and diversity of anomalous events, the cost of frame-level annotation, and the semantic complexity of real-world hostile activities. To address these limitations, we propose the Automated Surveillance and Hostile Intent Evaluation Network (ASHEN), a multi–modal weakly supervised framework that unifies vision–language representations with multi–modal fusion and temporal modeling. ASHEN combines CLIP ViT-B/32 features as a semantic backbone and augments them with YOLOv8 object detection and human–object interaction (HOI) descriptors through a Gated Multi–modal Fusion mechanism that learns adaptive softmax-weighted combinations of each modality. The proposed framework achieves state-of-the-art performance on both binary and multiclass video anomaly classification using only video-level labels. On the dedicated Ashen Dataset, ASHEN attains a binary AUC of 98.09%, a nine-class classification accuracy of 80.83%, and a Macro-AUC of 95.69%. These results indicate that ASHEN exhibits strong discriminative capability and suggests substantial promise for proactive threat identification and monitoring in critical infrastructure environments.

Keywords: Video Anomaly Detection, Weakly Supervised Learning, Multiple Instance Learning, Contrastive Language-Image Pretraining, Multi–modal Fusion, Temporal Transformer, Intelligent Surveillance

1. Introduction

The rapid expansion of surveillance camera systems across urban environments, transportation networks, and critical infrastructure has generated a massive volume of video data that far exceeds the capacity of human operators to monitor in real time. While the number of deployed cameras has surpassed hundreds of millions worldwide, the vast majority of recorded footage is never reviewed, rendering continuous manual monitoring both impractical and economically infeasible. This disparity has motivated significant research interest in automated Video Anomaly Detection (VAD) (1; 2; 3), which seeks to identify unusual, suspicious, or potentially dangerous events in video streams without exhaustive human oversight. VAD presents fundamental challenges: anomalous events are inherently rare, highly diverse in appearance and context, and often ambiguous in interpretation. An event considered anomalous in one setting—such as running in a hospital corridor—may be entirely normal in another, such as a sports arena. Consequently, developing robust and generalizable anomaly detection systems remains one of the most pressing open problems in computer vision and intelligent surveillance.

Early approaches to VAD relied on fully supervised learning, which requires dense frame-level annotations indicating the precise temporal extent of each anomalous event. While effective in controlled

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